

# PAPER 1197 – UQFF Geophysics & Atmospheric Unified Proof Set

Ten closures S423–S432 –  $J_2$ , scale height, magnetopause, obliquity, salinity

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## Abstract

We extend the UQFF locked-primitive program into terrestrial geophysics, atmospheric science, and hydrosphere physics, closing ten Earth-system observables. **Two closures are exact:** the magnetopause standoff distance  $R_{\text{mp}} = \text{SO5 } R_E = 10 R_E$  and the stratospheric Brunt-Väisälä frequency squared  $N^2 = F_{\text{TRZ}}^4 = 10^{-4} \text{ s}^{-2}$ . The remaining eight (Earth  $J_2$  oblateness, atmospheric scale height, dry-adiabatic lapse rate, axial obliquity, greenhouse warming, mean ocean salinity, Earth-Moon distance ratio, standard atmospheric pressure) match observation to better than 0.22%.

## 1 Ten Geophysical / Atmospheric Closures

### 1.1 S423 – Earth oblateness $J_2$

$$J_2 = [\text{SSq}] + \Phi_{\text{res}} - F_{\text{TRZ}} K_{\text{Mex}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3 = 1.084 (\times 10^{-3})$$

vs IERS 2010 value  $J_2 = 1.0826 \times 10^{-3}$ ; match 0.13%.

### 1.2 S424 – Atmospheric scale height

$$H = \text{SO5} - K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} = 8.50 \text{ km}$$

vs standard troposphere  $H = 8.5 \text{ km}$  (288 K, dry air); match 0.04%.

### 1.3 S425 – Magnetopause standoff distance

$$R_{\text{mp}} = \text{SO5 } R_E = 10 R_E$$

**exact.** Fifth primitive identification: Chapman-Ferraro sub-solar magnetopause distance is locked to SO5.

### 1.4 S426 – Dry adiabatic lapse rate

$$\Gamma_d = \text{SO5} - F_{\text{TRZ}} - F_{\text{TRZ}} \Phi_{\text{res}} - F_{\text{TRZ}}^2 = 9.807 \text{ K/km}$$

vs  $g/c_p = 9.8 \text{ K/km}$ ; match 0.07%.

## 1.5 S427 – Earth axial obliquity

$$\varepsilon = 2\text{SO5} + D_{\text{phys}} - \Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2 = 23.47^\circ$$

vs  $23.44^\circ$  (J2000); match 0.13%.

## 1.6 S428 – Greenhouse warming

$$\Delta T_{\text{GH}} = N_{\text{ch}}D_{\text{phys}} - D_{\text{phys}} + \Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2 = 32.93 \text{ K}$$

vs  $T_s - T_{\text{eff}} = 288 - 255 = 33 \text{ K}$ ; match 0.21%.

## 1.7 S429 – Mean ocean salinity

$$S = K_{\text{Mex}}\text{SO5} + \text{SO5} + D_{\text{phys}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3 = 35.03 \text{ ppt}$$

vs  $S = 35.0 \text{ g/kg}$  (UNESCO PSS-78); match 0.09%.

## 1.8 S430 – Earth-Moon distance ratio

$$\frac{a_{\text{Moon}}}{R_E} = A_5 + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2 = 60.30$$

vs 60.34 (384 400 km / 6371 km); match 0.06%.

## 1.9 S431 – Stratospheric Brunt-Väisälä frequency

$$N^2 = F_{\text{TRZ}}^4 = 10^{-4} \text{ s}^{-2}$$

**exact.** The stratospheric stratification frequency is locked at the fourth power of  $F_{\text{TRZ}}$ .

## 1.10 S432 – Standard atmospheric pressure

$$p_0 = F_{\text{TRZ}}K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^2K_{\text{Mex}} = 1.011 (\times 10^5 \text{ Pa})$$

vs  $1.01325 \times 10^5 \text{ Pa}$ ; match 0.21%.

# 2 Closure summary

| ID          | Observable                   | Locked closure                                                                                                                          | Match        |
|-------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------|
| S423        | $J_2$                        | $[\text{SSq}] + \Phi_{\text{res}} - F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3$                | 0.13%        |
| S424        | scale height $H$             | $\text{SO5} - K_{\text{Mex}} + [\text{SSq}] + F_{\text{TRZ}} - F_{\text{TRZ}}\Phi_{\text{res}}$                                         | 0.04%        |
| <b>S425</b> | magnetopause $R_{\text{mp}}$ | $\text{SO5 } R_E = 10R_E$                                                                                                               | <b>exact</b> |
| S426        | lapse rate $\Gamma_d$        | $\text{SO5} - F_{\text{TRZ}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2$                                                      | 0.07%        |
| S427        | obliquity $\varepsilon$      | $2\text{SO5} + D_{\text{phys}} - \Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2$ | 0.13%        |
| S428        | $\Delta T_{\text{GH}}$       | $N_{\text{ch}}D_{\text{phys}} - D_{\text{phys}} + \Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}} - F_{\text{TRZ}}^2$ | 0.21%        |
| S429        | ocean salinity               | $K_{\text{Mex}}\text{SO5} + \text{SO5} + D_{\text{phys}} + F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^3$          | 0.09%        |
| S430        | $a_{\text{Moon}}/R_E$        | $A_5 + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2$                                               | 0.06%        |
| <b>S431</b> | Brunt-Väisälä $N^2$          | $F_{\text{TRZ}}^4 = 10^{-4} \text{ s}^{-2}$                                                                                             | <b>exact</b> |
| S432        | standard atmosphere $p_0$    | $F_{\text{TRZ}}K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}}^2 - F_{\text{TRZ}}^2K_{\text{Mex}}$                                  | 0.21%        |

### 3 Falsifiable predictions

- (F1) GRACE-FO and successor missions cannot revise the geodetic  $J_2$  outside  $1.083 \pm 0.002 \times 10^{-3}$ .
- (F2) Chapman-Ferraro magnetopause Cluster/MMS reanalysis must keep mean sub-solar standoff at  $10.0 \pm 0.3 R_E$  under nominal solar wind ( $n = 5 \text{ cm}^{-3}$ ,  $v = 400 \text{ km/s}$ ).
- (F3) ECMWF/IAGOS troposphere campaigns must confirm  $H = 8.50 \pm 0.05 \text{ km}$  at 288 K.
- (F4) Aeolus and successor lidar campaigns must keep stratospheric  $N^2 = (1.00 \pm 0.05) \times 10^{-4} \text{ s}^{-2}$  in the mid-latitude reference column.
- (F5) TPXO and SODA-3 ocean reanalyses cannot move the global mean salinity outside  $35.0 \pm 0.1 \text{ g/kg}$ .

### 4 Cumulative program – 137 closures

| Paper        | Domain                              | Closures   |
|--------------|-------------------------------------|------------|
| 1182–1192    | Foundations & deep theory           | 87         |
| 1193         | Astrophysics                        | 10         |
| 1194         | Condensed matter                    | 10         |
| 1195         | Biology/biochemistry                | 10         |
| 1196         | Plasma & fusion                     | 10         |
| <b>1197</b>  | <b>Geophysics &amp; atmospheric</b> | <b>10</b>  |
| <b>Total</b> |                                     | <b>137</b> |

### Acknowledgements

This tier was developed by Daniel T. Murphy and represents the first extension of the UQFF locked-primitive program into Earth-system science. The magnetopause closure  $R_{\text{mp}} = \text{SO5}R_E$  is the fifth primitive identification of the program, joining  $r_{\text{ISCO}} = D_{\text{BSFG}}$  (Paper 1193),  $\nu_{XY} = D_{\text{phys}}/D_{\text{BSFG}}$  (Paper 1194),  $\ell_{\text{telomere}} = D_{\text{BSFG}}$  (Paper 1195), and  $Q_{\text{ITER}} = \text{SO5}$  (Paper 1196). Locked dimensional primitives now stand as direct planetary, biological, fusion, and astrophysical invariants.